HYDRIS AI

HYDRIS FIELD GLASSES · PRODUCT BOOK, PART 11

Roadmap and future products

Roadmap and future products

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Document control

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1 Roadmap and future products

1.1 Purpose of the roadmap

The minimum viable product is built and runs as a zero-configuration demo. The next twelve months turn it into something a plant can rely on every shift, then into something a regulator, a maintenance system and a control room can rely on too. This part is a plan, not a commitment: every date is an estimate that assumes the current team and no major change in Meta's developer programme, and effort figures are team estimates. Status labels follow the book: **Built**, **Partial**, **Planned**.

5
quarters planned
Q4 2026 to Q4 2027

8
themes
from hardening to ecosystem

19
future features analysed
problem, outcome, dependencies

1 to 2
pilot plants
ETP or STP, Q1 2027 (target)

1.2 Horizons: now, next, later

Table 1. Horizons, goals and exit conditions (plan)

Horizon	Window	Goal	Exit condition
Now	Q4 2026 to Q1 2027	A pilot plant uses Hydris every shift without engineering support on site	Pilot exit criteria all checked; 4 weeks of daily use
Next	Q2 to Q3 2027	A customer can buy, deploy and audit Hydris without Hydris engineers	GA release; SOC 2 Type I report; Play or MDM distribution; second paying site
Later	Q4 2027 and beyond	Hydris is the system of record for field work and compliance evidence at multi-site customers	A regulator accepts a Hydris compliance report; one CMMS integration live

Confidence falls with distance: scope is high-confidence for Now, medium for Next and low for Later; dates are medium, low and very low. Pilot customer dates, Meta DAT access, pilot KPIs and team size are what move them.

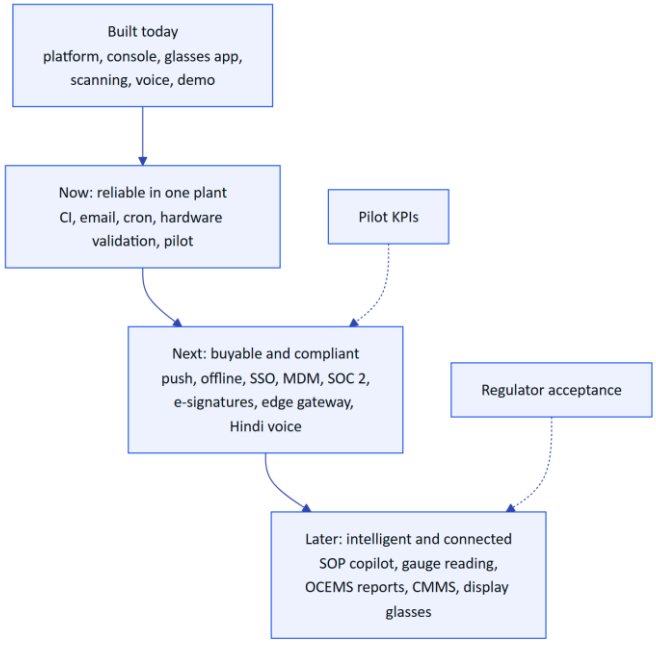


Figure 1. *From MVP to system of record: what each horizon unlocks (plan)*

1.3 Timeline, Q4 2026 to Q4 2027

The timeline and Gantt chart are the plan as of October 2026 and are re-baselined at the end of every quarter.

Table 2. Quarter summary (plan)

Quarter	Headline	Key deliverables	Exit signal
Q4 2026	Make it reliable	CI, email transport, cron schedule, audit images to object storage, purge jobs, Android pilot build	Test gates green on every change
Q1 2027	Prove it in a plant	Ray-Ban Meta hardware matrix, pilot plant 1, field trial, KPI baseline	Pilot exit criteria met
Q2 2027	Make it buyable	GA: push notifications, offline outbox, SSO, MDM, SOC 2 Type I readiness, India-region production	Second site signed
Q3 2027	Make it compliant and connected	E-signatures, edge gateway, Hindi voice	First e-signed record approved at a customer
Q4 2027	Make it intelligent	SOP copilot, gauge reading, first CMMS connector, OCEMS report drafts	Copilot used weekly by 50 % of operators (target)

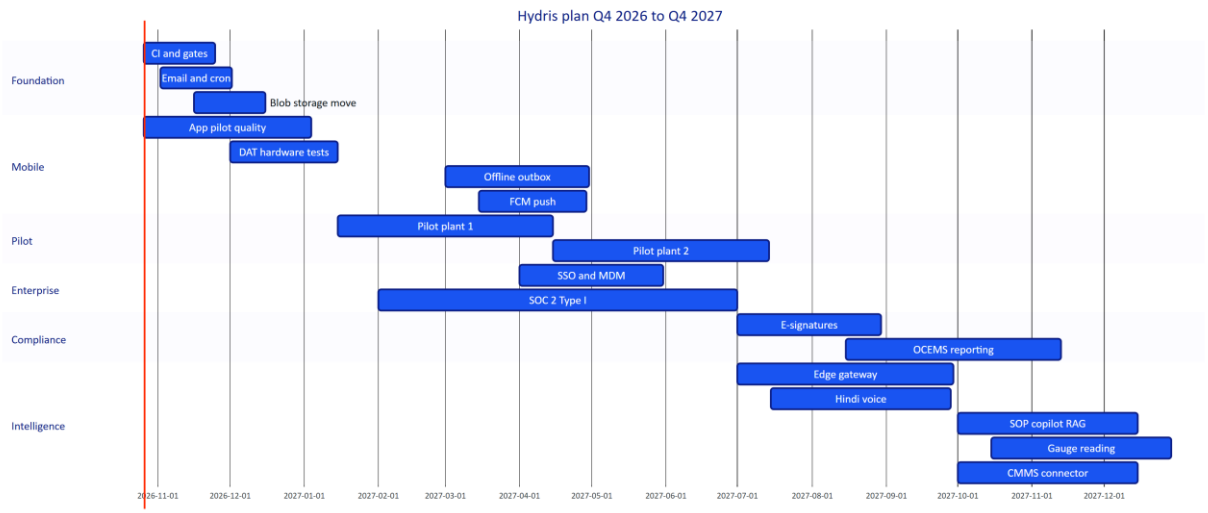


Figure 2. *Quarter plan as a Gantt chart (plan; bars are estimates)*

1.4 Known gaps to close first

These are the debts that would hurt a pilot. They come before any new product.

Table 3. Known gaps and tech debt, in priority order

Gap	Impact	Fix	Target
No CI pipeline	Regressions reach the demo	Typecheck, unit, API and e2e gates on every change (NFR-010)	Q4 2026
Email not sent	Invitations and resets never arrive	Email transport for the outbox (FR-086)	Q4 2026
Cron runs daily on Vercel	Stale-data alarms evaluated once a day (FR-035)	Per-minute schedule or external scheduler	Q4 2026
Audit images stored in Postgres	Database bloat, slow backups	Object storage with lifecycle	Q4 2026

Gap	Impact	Fix	Target
No purge jobs	Expired sessions, challenges and sent mail kept indefinitely	Purge 30 days after expiry	Q4 2026
No Android automated tests	Manual regression only	Android test suite	Q4 2026
DAT paths unverified on hardware	Glasses camera, button and microphone may fail on device (FR-055 to FR-057, FR-074)	Hardware test matrix	Q1 2027
Web glasses refresh token in browser storage	Token theft on a compromised device	Native app keystore (built); shorter or bound tokens for web	Q1 2027
No per-org opt-out of third-party AI	Blocks some customers	Organization setting	Q2 2027
No device attestation	Spoofed clients	Play Integrity	Q2 2027
Audit log not hash-chained	Weaker tamper evidence	Hash chain or WORM export	Q3 2027

1.5 Future products in brief

Each future product reuses a pattern already proven in the MVP: read-only and outbound-only towards the plant, machines propose and people confirm, and every value keeps its provenance.

Table 4. Future products (all Planned; effort is a team estimate)

Product	Problem it solves	Outcome	When (plan)	Effort
Edge gateway (OPC UA, MQTT, Modbus)	Each customer needs custom integration; Hydris accepts HTTPS batches only	A container in the plant network reads tags and forwards readings outbound over HTTPS, with store-and-forward and a tag-mapping UI (FR-027, FR-028)	Q3 2027	RICE effort score 5 (estimate)
Offline-first sync	Plants have dead zones; only glasses inspections queue today	Android outbox with idempotent replay for acknowledgements and checklist ticks (FR-098)	Q2 2027	3 person-months
Predictive maintenance	Pumps and blowers fail with warning signs nobody watches	Per-asset baselines flag drift and create advisory maintenance tasks with evidence	Later	6 person-months
SOP copilot (RAG)	Operators ask "how" and "why" questions that need long procedures	Answers from approved documents only, with citations, never inventing a step	Q4 2027	3 person-months
Multilingual voice	Operators speak Hindi, Tamil, Marathi, Gujarati	Same grounding guarantees in each language (FR-073): Hindi first, then Tamil and Marathi	Q3 2027 onwards	4 person-months
Gauge reading by computer vision	Analogue gauges are read by eye and copied	"Read this": Hydris proposes a value with confidence, the operator confirms, the photo is kept	Q4 2027	5 person-months

Product	Problem it solves	Outcome	When (plan)	Effort
Compliance reporting and OCEMS	Plants assemble CPCB and SPCB returns by hand	Evidence assembled with provenance, exceedances and data availability, draft returns for e-signed approval (FR-120)	Q3 to Q4 2027	4 person-months
E-signature	Approval has no signature manifestation	Re-authentication at signing, stated meaning, name and time bound to the record (FR-068)	Q3 2027	2 person-months
Display glasses	Ray-Ban Meta has no display, so operators look down at the phone	The existing heads-up layout renders on display glasses and industrial headsets behind a display adapter	Later	3 to 5 person-months per target
iOS app	Some supervisors and managers carry iPhones	Supervisor companion first (alarms, reviews, approvals)	Later	6 person-months
CMMS integrations	Hydris tasks would duplicate SAP PM or IBM Maximo work orders	Two-way sync of work orders and assets, plus webhooks for LIMS, ERP and ticketing	Q4 2027 onwards	4 person-months for the first connector

Edge gateway and the compliance pipeline

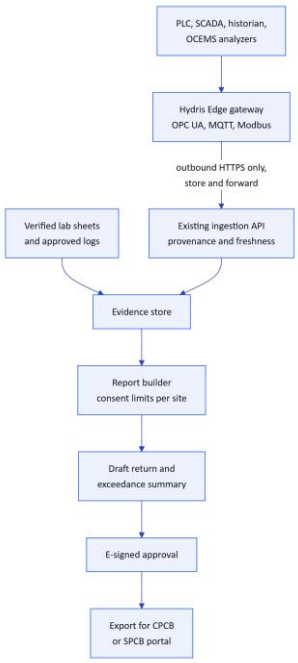


Figure 3. *How the edge gateway and compliance reporting extend today's ingestion (planned)*

The gateway reuses the single ingestion path, so readings from industrial protocols carry the same guarantees as today's HTTPS batches. Compliance reporting then builds on verified scans and e-signatures; direct submission to regulator systems happens only where an official interface exists.

Display glasses and new inputs

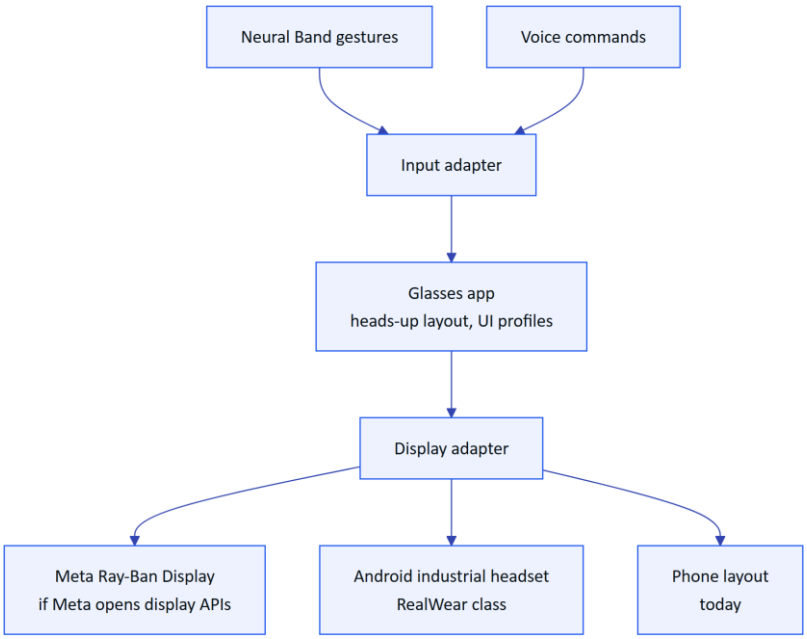


Figure 4. *Display adapter for future display glasses and headsets (planned)*

The heads-up layout and input abstraction already exist; display targets plug in behind adapters. The dependency is Meta exposing display and Neural Band input to third parties, which is not available today.

1.6 Value versus effort

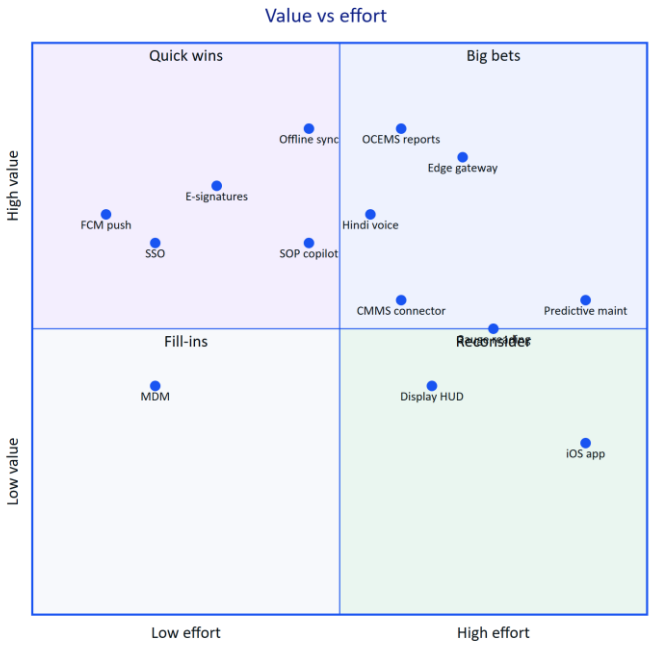


Figure 5. *Value versus effort for candidate epics (team estimates)*

Quick wins (push, SSO, e-signatures) land first because they unblock buying. Offline sync, OCEMS reports and the edge gateway are the big bets that make Hydris a system of record. iOS and predictive maintenance wait for evidence of demand and enough labelled history. A RICE score backs this ordering: push notifications score highest (270), then SSO (180) and offline sync (160); predictive maintenance scores lowest (5).

RISK

The biggest risks are a small team spread across too many surfaces, pilot customer delays (approvals, camera policy) and Meta DAT publishing staying gated for India-based developers. Mitigations: strict horizon discipline (Later items stay unstarted until GA), a site camera-policy pack and a second pilot candidate, and the phone-camera path, which works without DAT.

1.7 Success metrics per horizon

Every figure below is a target.

Table 5. Success metrics per horizon (targets)

Horizon	Metric	Target
Now	Pilot exit criteria complete	100 %
Now	Rounds completed on time at the pilot	95 %
Now	Median alarm acknowledgement, critical and high	Under 5 minutes
Now	Audit fields where independent engines agree	90 %
Next	Weekly active operators over licensed operators	70 %
Next	Paying sites	3
Next	Offline actions synced without duplicates	100 %
Next	SOC 2 Type I	Report issued
Later	Compliance reports generated in Hydris	Accepted by 1 regulator
Later	Copilot answers with citations	90 % of answered questions
Later	Multi-site customers	2

1.8 Release cadence

API and web ship continuously to staging and weekly to production after the gates, with S1 hotfixes within 24 hours. The Android app ships every two weeks through Play internal, closed and managed tracks, with glasses features behind remote flags. The edge gateway will ship monthly as signed container images inside a customer-controlled upgrade window, and this book is updated with each release.